Lecture 8: Conscious of the Present; Conscious of the Past: Vision and Memory (cont.) (full transcript)

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Overview: In this lecture, Professor Bloom reviews the basic psychological research on memory. Specific topics covered include the different memory types, memory limitations, strategies that improve memory, and memory disorders. This lecture also includes a discussion of several important social implications for memory research, such as recovered memories, and the influence of suggestibility on eyewitness testimony.

Chapter 1. Distinctions between Short-Term and Long-Term Memory [00:00:00]

Professor Paul Bloom: I'll begin the class officially with a different sort of demonstration. I want to just show you one of the change-blindness studies that has been done in the real world. And these videotapes are not available publicly. We get them from the web and see them as little Java scripts. So, this is one of the first studies done by Dan Simons when he was at Cornell. And his adviser at the time was our Frank Keil, who's now in our department. So, here's the study. [The videotape shows a student who is walking outside on a college campus and stops and asks an older man for directions. Their conversation is interrupted when two construction workers carrying a door walk in between them. At that point the student surreptitiously swaps places with one of the men carrying the door. Remarkably, the conversation about directions resumes without the older man ever noticing that he is speaking to a completely different person.] [laughter] And you don't notice it.

Change blindness is one of the more striking phenomena discovered by laboratory scientists and by psychologists. But it's important to realize, to get away from the sort of surprise of the gorilla and the fact that it's hard to see the flickering — the object that's flickering, and appreciate the big moral of this, because the big moral of this is actually, I think, striking and quite important. You think right now that you're perceiving the world. I look down on you and I think I have a whole sense of where everybody is. I can't see everybody perfectly in back. You're kind of far away and blurry but there's a sense in which I have a world around me. Similarly, if I'm to close my eyes for a second, everything just remains and I could sort of remember some of the things that are there. That's really good sound localization by me [responding to a noise he heard from the video screen].

So you're looking up and you think you have a sense of the world both in perception and memory. The change-blindness experiment suggested this isn't true. The change-blindness experiment suggests that if you look at me for a second and during that second all of your classmates change positions, including those next to you, you are extremely unlikely to notice. The change-blindness experiment suggests that if you turn your eyes away from me towards there for a second and turn back, and I'm dressed entirely differently, you wouldn't notice. The exceptions would be if you told yourself consciously, "Remember what this guy is wearing; he's wearing this, that and the other." But if you don't do it consciously you'll lose it, and usually this is okay.

Usually, it's okay because your memory and your visual system exploits a basic fact about the universe, which is that most things stay the same most of the time. I don't have to explicitly remember that

you're over there when I turn my head for a second because you'll be over there in any case. You don't need to hold precise representations of the world. And so you only notice it in certain clever circumstances. One sort of clever circumstance is when psychologists change reality as in the change-blindness studies. A second sort of circumstance is in movies. So, one of the big surprises when people started making movies involving cuts was it is extremely difficult to get everything continuously right. And you need to work very hard to notice. So, there's all of these continuity errors that creep up into movies and you have to be a film buff or writing it down to even notice this. And the overall moral here then is that your perception of reality is a lot more sparse, a lot more limited, than you might think it is.

So, this is where we were at the end of last class. We were talking about the different sorts of memories: Sensory memory, which is the sort of fraction of a second of sensory residue of what you're hearing and what you're seeing, working memory, short-term memory, and then long-term memory. And we talked last class about how things get into sensory memory, into working memory, the role of attention. And in fact, the change-blindness studies are actually just studies of how something gets from your senses to your consciousness and what does and what doesn't.

Now I want to move to the distinction between working memory, short-term memory, and long-term memory. Now, the obvious distinction is actually just in fact — is storage differences. So, long-term memory or "LTM" has a huge storage capacity. This is your memory like the hard drive of your computer. This is the memory you walk around with. It includes all the words in English, just for example, 60 to 80,000 words. It includes everybody you've ever met, languages, faces, stories, locations, nursery rhymes, songs, TV programs. Nobody knows the storage. It is not true that you remember everything that has ever happened to you. There's no reason to believe that this is true. At the same time though, you have a huge amount stored in your brain in long-term storage and nobody actually — It has to be limited because it's a finite, limited brain. But nobody knows how big it is. Nobody knows how many terabytes you carry around in your brain and — but it's a lot.

Compare this to working memory – the short-term memory, which is actually very limited. Your memory of what you could store on — in — where you could hold in consciousness right now is quite limited. Here is an exercise. Do not write these things down. I want you to remember them. I'm just going to give you a few numbers: 14, 59, 11, 109, 43, 58, 98, 487, 25, 389, 54. Please write them down. View this as an IQ test if that would relax you. How many of you who decided to participate in this experiment got three or less? Good. Good. Four, five, six, seven, eight, nine or more? Anybody get all eleven? This is a particularly difficult memory task. The numbers are meaningless. And I told — and I forgot to tell you to get your pen and pencil ready, so some of you just glared at me. But [laughter] under normal circumstances the cognitive psychologist George Miller said that this sort of suggested that the standard memory storage of short-term memory is seven, plus or minus two. And what that means is anywhere from five to nine roughly.

Some of you, I bet, can beat that. Some of you on a not-so-good day maybe won't make it that much. Now "seven plus or minus two" is what you — ;so, that's what you hold in consciousness. I can tell you 14, 21. You walk around, "Oh, yeah, 14, 21." You hold that in consciousness with no problem. But I throw eleven numbers at you, you can't. Some dribble out. You can't hold that in your conscious window in your short-term memory.

Now, this raises the question "seven plus or minus two" what? And the answer seems to be what George Miller calls "chunks." And a chunk is a basic memory unit, something you think of as a single, individual entity. So, suppose you see the string of letters "L, A, M, A, I, S, O, N." If you don't know — If you can't form these into words and you have to remember them, these are eight chunks. You have to just pick them up separately. On the other hand, if you break them up into four words you could just remember it as four chunks. And if you break it up into two words in French, "la maison," "the house," it could just be one or two. How much you know depends — affects how much you memorize — how much you could store in memory because it affects what counts as a basic unit of memory.

And there's all sorts of examples of this. If I tell you "1, 1, 0, 1, 1, 0, 0, 1, 0, 1, 1, 0," those of you who don't know binary numbers might have to remember that as "1, 1, 0, 0," whatever I said. Those of you who are computer scientists or mathematicians or, for whatever reason, know binary numbers could convert it into a single binary number. Anybody know what the number is? No, I cannot say it again. [laughter] Some number, 24, or not 24 — to some number, 24, and then you remember "24." It's easier. Suppose you see a chessboard and the chessboard is set up and you don't know how to play chess. It is murderously hard to remember that. They've done the experiments. They've taken people in a lab who don't know how to play chess. They set up a chessboard and then they say, "Okay. Look at this for five minutes." Then they take it away, set it up again, and it's murderously hard. "There is a horse-y thing on the side there and everything." But if these chess pieces are set up in some way that's logical for a chess player, then a chess master could look at it and remember it in a glance, "Oh. It's the Fibonacci defense" or something like that [laughs], and then immediately recover it.

Similarly, football coaches have been tested on their memories of football diagrams. And they have a photographic memory for football diagrams because it corresponds to things that make sense. Architects could have a photographic memory, a perfect memory for floor plans because it makes sense to them. They understand it. And so the way you store things in memory, and this is a theme we're going to return to when we get to long-term memory, depends in a large extent on how much you understand it. And this shows up in expertise effects.

Now, this is what's happening so far in short-term memory, how much you hold in there. The question is how do you get it into long-term memory? So, you have long-term memory, your major storage system. How does information get from your consciousness to long-term storage? Well, there's one thing — there's one way which sort of works sometimes but not very well. And it's called "maintenance rehearsal."

Suppose I said you have to remember this number, this string of numbers. And if you remember it in twenty minutes you will get one thousand dollars. And the string is my phone number when I was a kid. I'll include the area code: 514-688-9057. Now, if you tell that to a four-year-old, well, the four-year-old will say, "I'll remember it." And then you ask them, "What did I just say?" "Well, I don't know." If you tell it to an — because you know something — If a lot depended on it, you would know to do something. What you would do is you'd say to yourself, "514-688-9057, 514-688-9057, 514-688 — " You'd rehearse it in your head over and over again. The problem is you could hold it as long as you can do that. It's like these movies. You see this all the time, like an episode of 24: "Jack, call CTU and tell them Agent 11 is trapped in a — " And I can't even remember this but the way to remember it is you

hold — you've just got to repeat it over and over again in your head. But this will not typically get things into long-term memory.

Chapter 2. How Structure and Organization Affects Long-Term Memory [00:13:22]

To get things into long-term memory, rehearsal is usually not enough. You need to do other things. Typically, what you need is structure and organization. And one way to demonstrate this was in a classic "depth of processing" experiment which nicely illustrates the fact that the more you structure something, the deeper you think about it, the better it gets entrenched in the long-term memory.

So, in this study what they did was they asked people — they told people that there's going to be words flashed on a screen. And all of the subjects saw the same strings of words. There were forty-eight words. They were not told to memorize the words. One third of the subjects was told, "Look. Some of these words are going to come out in capital letters, some of them not capital letters. Press a button for capitals, non-capitals." "Sure." The other group was told, "Some of these words will rhyme with 'train,' Others won't. Press a button if it rhymes in 'train'." The third group was told, "Does it fit into the sentence ‘The girl placed the blank on the table'? Press a button if it does. Press a button if it doesn't." Then they were asked as a surprise, "What words did you see?" And the findings looked like this. When they were asked to focus on just what the word looked like, memory was very poor, the sound better, the meaning better. If you want to remember something, the best way to remember it is to give it meaning, to give it sense.

This is illustrated through a very ancient technique, which is that the way to remember things that are otherwise arbitrary is to give them some organization through memory tricks, through vivid imagery or songs or poetry. And there's a lot of examples of this. Do you know how to remember that the hippocampus — There's a part of the brain called the hippocampus. This is the worst memory trick ever but it will stick with you for twenty years. The hippocampus is involved in spatial memory. It's involved in finding your way around. Think to yourself, "The way I find my way around campus is through the hippocampus." And you think, "Well, that's stupid," but you'll never forget now that the hippocampus is in charge of spatial memory. It's going to be all you retain from this course.

Memory books on how to remember people's names usually try to exploit this sort of thing when you try to get poetry or dramatic images. So, the memory books always typically involve somebody — like you meet somebody with very spiky hair and they say, "My name is Mr. Fish" and then you remember — you think of their — of a big fish impaled on their hair. And then whenever you see them you remember their name. It only really works for names like "Fish" but [laughter] the idea is you try to generate vivid imagery. When stuck with a situation where you have to remember ten letters, turn it into a song where — or a dirty poem where each of the letters is the first words of it. When having to remember something that seems totally arbitrary, try to figure out a grand and obscene image that will come to mind easily. And this is how — these are one way to get things into memory.

At a deep level though, the way to get things into memory, and this applies to this course no less than anything else, is by understanding the — understanding it. I'm going to read you something and I want you to try your best to remember what I tell you. These are not going to be strings of numbers. These are going to be — This is going to be a series of sentences: "A newspaper is better than a magazine. A seashore is a better place than the street. At first it is better to run than to walk. You may have to try

several times. It takes some skill but it's easy to learn. Even young children can enjoy it. Once successful, complications are minimal. Birds seldom get too close. Rain, however, soaks in very fast. Too many people doing the same things can also cause problems. One needs lots of room. If there are no complications, it can be very peaceful. Finally, a rock will serve as an anchor. If things break loose from it, however, you will not have a second chance."

And here is what I said [pointing to the written sentences on the overhead slide]. This is murderously hard to remember. Now try it. [Pointing to a slide that reads, "This paragraph is about flying a kite."] Knowing what this is about, being able to put a context to it helps the memory and helps it come to mind. [laughter] Okay.

Chapter 3. Memory Retrieval [00:18:29]

So, this is about how to get memory — how to get information into your memory. How do you get information out? So, it's exam period. You got the stuff presumably into your head. You have to get it out. You have to retrieve it. There is a court case. You have to figure out — You have to recount the crime that you witnessed. You see somebody and you want to know his or her name. And you heard it; you just have to get it out. Well, how do you do that? Well, there's "retrieval cues." Retrieval cues make sense. Retrieval cues are just things that have been associated with what you — what you're trying to remember. If I have to remember to replace the windows, when I walk in to my living room and see that a window is cracked that will remind me to replace the windows. If I had a lunch date with you and forgot about it, when I see you, "Oh, yeah. We were supposed to get together to have lunch."

Retrieval cues bring things back but it's a little bit more complicated than that. There's a more general relationship between encoding and retrieval called the "compatibility principle." And what this means is you're much better to remember something in the context in which you have learned it. And this is also known as "context-dependent memory" and "state-dependent memory." It's illustrated by one of the strangest experiments in the history of psychology where they had people on a boat and then they had them scuba dive underwater. And they taught them things either on the boat or underwater with things that they held up. And then they tested them later. And it turns out that you'll remember it better if you're tested on it in the context in which you learned it. And it might be because then the retrieval cues help bring it back. But it's more general than that. If you have to remember something you learned in this class, you will do better if you try to think about the room in which you learned it in. You will do better on your final exam if you were to take it in this room than if you were to take it in another room because being in this room will bring back the cues.

It's not just the environment. People who learn things when they're stoned remember them better — keeping stoned at a sort of a low-level that doesn't disrupt other mental activities — remember them better when they're sort of stoned again [laughter] than if they're non-stoned. Similarly — So, if you study while you drink you should tipple a little bit before coming in to the Final exam. [laughter] It's sort of like the "what happens in Vegas stays in Vegas" sort of result. And so, similarly, it even applies to moods in that if you learn something when depressed you have a slightly better recovery of it when you're in that same mood of depression than when you're elated. And the idea is that part of what memory is — part of what recovering memory is is getting back your original context in which you learned it.

"Elaborative rehearsal" and retrieval involves the connections between different things. Elaborative rehearsal is that the more you think about something the easier it is to remember. If you have to think about — If you have to remember something, try to connect it to as many things as possible. Think of an image. Make a joke out of it. Imagine how you would explain it to somebody else. Imagine how the world — what the world would be like if it wasn't so. And the idea is that this sort of thinking about it makes connections in your memory from that thing you have to learn, to other memories. And so it makes it easier to recover.

"Elaborative retrieval" refers to a finding that when you want to get something back out of memory people tend to give up too soon. It turns out that there's a lot of stuff that's in your memory but it needs work to extract; it needs various sort of searching strategies. One study asked people who were considerably older than you to remember their high school classmates. And in the first pass people were terrible. Maybe they had a couple of friends they kept in touch with. Otherwise, pretty bad. And this is a good experiment because you could use high school yearbooks to judge whether or not they get it right. But then what you do is you tell the person, "Look. Keep trying. Were you — What sort of — Who was your teacher? What sort of clubs did you belong to? What sort of sports you — did you participate in? How did you get to school? How did you get from school? What did you do during lunch? What did you do during break?" And you keep ask — "Do you know — have any friends whose letter — whose last name began with ‘B,' with ‘C,' with ‘D'?" And you keep pushing and pushing and pushing. And over the span of time things come back. Again, it's not true that you never forget. There is honest to God forgetting but sometimes you think you forget and it's because you haven't looked long enough. There's a real physical notion of searching for the right answer.

We've talked about retrieval. Oh. Every class I've given somebody asks either in class or by e-mail what about déjà vu? And déjà vu is a feeling that an event has happened before. So, you're looking at me and I'm lecturing and you say, "I've heard this before. I know this before." You see somebody and say, "I've been in this situation before." This is not evidence for psychic powers, [laughter] which many people say it is, but nobody really knows why this exists. We know, and this is a clue, it's worse with frontal lobe damage. If you get damage to this part of the brain, you get a lot more déjà vu experiences. I asked some experts in memory, including Marcia Johnson who is chair of our department, what the best explanation for déjà vu is. And the answer she gave, the — say one big theory, goes like this. Déjà vu is a feeling that it's happened before. The answer is it has happened before. It's happened half a second ago. And so what happens is sometimes there is a glitch, a disturbance in the force. I don't know. There's a glitch [laughter] and you are talking and then something happens to you and you put it in your memory. But it's as if you don't put the stamp on it of what time and what date. So, you're talking to me and then you store it in memory but you don't store it in memory as happening right now. Then half a second goes by and you're talking to me and you say, "This is strangely familiar." And that's one theory of what goes on in déjà vu.

Chapter 4. Memory Failure [00:25:51]

Okay. So far, there's the sort of good news – remembering – but then there's bad news – forgetting. How many people can remember, without looking down at your notes, at least two of the numbers I gave you earlier? How many people can remember at least four? Oh, impressive. If I asked you in an

hour, the number would go down. These are sort of statistics in a similar experiment [pointing to an overhead slide]. And this graph illustrates that people forget. Over time, you'll forget.

Why do you forget? Why is there forgetting at all? Well, there's different explanations for this. One explanation is your brain's a physical thing, it's a physical piece of meat, and it kind of goes bad. Physical things decay. And so, the memory traces that are laid onto your brain will just decay over time. A second answer is interference. So, remember those numbers? Here's a few more: 114, 81, 66, 42. Well, the more information that comes in that's similar to the stuff you're trying to remember, it blocks your recovery of original information. So your ability to remember something can be impaired by learning more things which are related to it because they get confused in memory. Finally, and maybe this is most interesting, there are changes in retrieval cues. So, the more time goes by the more the world changes. And if your memory is to some extent dependent on cues bringing it back to life, then the change in retrieval cues can make it more difficult to recall certain things.

This leads to a puzzle where there's considerable scientific debate over the case of childhood amnesia. And the case of childhood amnesia is — doesn't refer to when a child gets brain damage and gets amnesia. What it refers to is people have a difficult time recovering very early memories. I want people to just take a second and try to think back on what your first memory is and roughly how old you were. How many people don't think you have a first memory until you were about five years old or older? Okay. How many people think you have the first memory of around age four or younger? How many people think you have the first memory of around age three or younger? Two or younger? How many of you think you have the first memory when you were about one years old or younger? And I'm not asking about past lives but that [laughs] happened last year. How old is your — roughly your first memory do you think? [pointing towards a student] How old?

Student: Between one and two.

Professor Paul Bloom: Between one and two? Anybody think they could beat that? Same guy? Yeah.

Student: One.

Professor Paul Bloom: One. [laughter] Anybody else? The literature is unclear on this because it's very difficult to test people's recollections of their first memories. If I'm to ask people about their first memories, they'll often say, "Oh, yeah. I remember I was in this room and there was a crib and I'm going ‘Ga ga, goo goo' [laughter] and I was on the potty. I was walking. I was so cute. I remember it." It's very difficult to tell and, as we'll discuss in some detail, there are a lot of reasons to distrust people when they — not that they're lying but to distrust the accuracy of people's memories.

We also know from studies about trauma where people have terrible experiences when they're one or two. Typically, this trauma is not remembered later on. People know of trauma because they're told about it but they don't typically remember it with any accuracy. Even children — older children don't remember back beyond that age. Nobody knows why childhood amnesia occurs. Nobody knows why it's very difficult to recover memories before about the age of three. One theory is that the retrieval cues change radically. I had a friend of mine who's a clinical psychologist and he suggested a new form of therapy where they make these giant tables and chairs and then they bring you in to the office and you're standing there with these giant tables and chairs [laughter] and all these memories of being a

baby would come flooding back. [laughter] And he dropped out of the field and — [laughter] Really, but it's such a cool idea.

Some people think language is to blame. So a child, a baby, starts out with no spoken or signed language. Language comes to be learned at around one, two, and three, and it might be that the learning of a language reformats your memory. And once the memory is reformatted it can't go back to the previous state prior to language in the end. It could be neural maturation. It could be that those memory parts of the brains grow around age two or three that just weren't there prior to that. And nobody really knows. It's a fascinating research area why — about memory changes early on.

Another case of memory failure is brain damage. And brain damage comes in a couple of flavors. There is retrograde amnesia; "retro" for past. Retrograde amnesia is when you lose some memory of the past. This could be in a case where you get some sort of head trauma and you lose memory of your entire episodic memory. But typically, if you have any sort of serious accident that involves you losing consciousness you'll have a blackout of some period prior to that, say, blow to the head. And the reason for this is as you're having these experiences now they need to kind of get consolidated into your brain. Your brain needs to rewire and catch up to the experiences you're having. A sudden blow to the head will knock you unconscious and then the memories that have happened immediately prior will not get consolidated and they'll be lost forever.

Another sort of memory is anterograde amnesia and this was the case of — This happens in Korsakoff's syndrome. It happens to a very famous patient known as H.M. who actually lives in Hartford, Connecticut. And it happened to Clive Wearing, the film you saw last class. And this sort of amnesia is a sort of amnesia where you lose the ability to form new memories. And so you live in a perpetual present, unable to accumulate new memories.

But it's actually a little bit more complicated than that. What happens is — And this was an exciting discovery about these patients that led to some real insights about normal memory — What happens is — And this is the brain damage in these cases, the temporal lobe and the hippocampus, very useful for spatial memory you'll know. One discovery made about people who couldn't form new memories is that they could form new memories, but of certain types. So for example, this is a task here involving filling in a star while looking in to a mirror. And if I asked you to do it you'd find it pretty difficult. It's just kind of difficult to do. You'd be clumsy at it. You bring in an amnesic who can't form new memories and you say, "Hey. I want you to try something new. I want you to try this star game." He'd say, "Okay. I've never seen it before but I'll do it." Tries it. Does very badly. You bring him in and over and over again — Each time he does it he starts off by saying, "I've never seen this before. I'll — I'm sure I'll give it a try" but he gets better and better at it. And this is known as implicit memory.

The claim is that in these sorts of cases you lose the abilities to form explicit conscious memories that you're aware of, that you understand. But some sorts of memories persist and you are able to form them. This has actually been illustrated in a couple of dramatic movies, one of them a very bad dramatic movie [laughs] where Drew Barrymore loses the ability to form new memories and somehow falls in love with Adam Sandler. [laughs] Definitely don't watch that. But a very good movie called "Memento," which is about a character who loses his ability to form new memories while trying to track down his wife's killer. "Memento" is a movie which is fascinating because it's told backwards. But throughout "Memento" there's another story told forwards. And I like this story because it very

dramatically illustrates what does, and what is and is not impaired in cases of severe memory damage. So, I'm going to show you a couple of clips that illustrate the disassociation from "Memento." [clip playing]

Now, the next scene is actually modeled after real experiments. [clip playing] Those of you who have seen the movie know that this ends up quite tragically for Sammy. I highly recommend the movie. We've dealt right now with two sorts of failures of memory. One is everyday failure of memory when you forget. How many of you remember three or more of the numbers I originally presented? Yeah? Go ahead.

Student: [inaudible]

Professor Paul Bloom: Fourteen, 59, 11. Is that right? [laughs] Fine. [laughter] All right. I'm going to ask you again in a month. [laughter] Well, people are supposed to forget [laughter] and some things will — you will forget. That's normal forgetting.

A second case is forgetting due to brain damage. Forgetting due to brain damage is exotic and unusual but it's interesting in that it illustrates some more general themes about how the mind works. Remember one theme of this course is we're going to look at exotic cases like the case of Clive Wearing, not just because they're interesting in their own right but sometimes by looking at the extremes we could learn something about how normal people's normal, intact minds and brains work.

Chapter 5. The Power of Suggestion on Memory [00:37:15]

The third case of forgetting is more interesting and it actually — Well, I want to do a little trial here. What I want to do is I want to — You to listen to three children describe an event that happened. I want you to come to some — your own guess. Imagine you were a judge, you were a childcare worker, you wanted to see — I want you to be — come to your own guess about who you believe and what you think happened. [audio playing] [inaudible] [laughter]

You've heard three children. Who do you believe? Who believes — There's three of them, one, two, three. Who believes the first one? Who believes the second one? Who believes the third one? Sort of an even split. Twenty-three hundred experts were shown these films and asked about the different actions, whether or not the person ripped the book, messed up the bear, tossed the book in the air and, as you could see, the majority thought that he did. This is work done by Steve Ceci who was gracious enough to lend me the film to use for teaching purposes.

It turns out the second girl was right. Absolutely nothing happened. [laughter] The teacher said, "There is somebody named Sam Stone who's going to come in." A guy walks in and says, "Hi," walks around and leaves. [laughter] The first and third children had their memories implanted, not through any sort of science fiction means. They had their memories implanted — Well, they had their memories implanted like this. Some of the children would just ask questions. The interviewer, by the way, was herself unaware of what happened so the interviewer was a perfectly naïve interviewer. And it turns out if you just interview children and you ask them questions about whether the book was ripped, "Did you see him? Did he really do it?" they don't say anything. They didn't see anything and they won't say anything.

Other children were told about Sam Stone. They were told a stereotype about Sam Stone – that he's very clumsy and he tends to rip things and he trips and he breaks things and he spills things. And in fact, the third child mentioned that in passing. He said, "He always does that." Just knowing this about Sam Stone tends to raise the proportion of kids who say, for instance, that he ripped the book.

Other children were given suggestions. They were given suggestive questioning. They were a series of leading questions like, "Oh. Sam Stone came in? Did he rip a book while he was there?" And still more children got both. And in fact, the children you saw were from this group. They heard Sam Stone being described as a clumsy fellow and they were given a series of suggestive questionings. In this condition they were given several suggestive questionings over the period of several months. These children, like the first child and the third child, are not lying. They honestly believe that Sam Stone came in and did these things. Also they believe it and they're so convincing in their belief that experts, including police officers and child caseworkers and judges and lawyers, find these children to be extremely believable. And I think they probably find them to be extremely believable because the children are not lying. They really believe they saw what they saw. But these memories were implanted. And Ceci, and many other investigators, study how memories can be implanted in people's minds through suggestion and through leading questions.

It turns out that the same sort of experiments and the same sort of research has been done with considerable success in implanting false memories in adults. There are dramatic cases of people remembering terrible crimes and confessing to them when actually, they didn't commit them. And this is not because they are lying. It's not even because they're, in some obvious sense, deranged or schizophrenic or delusional. Rather, they have persuaded themselves, or more often been persuaded by others, that these things have actually happened.

Psychologists have studied in the laboratory how one could do this, how one can implant memories in other people. And some things are sort of standard. Suppose I was to tell you a story about a trip I took to the dentist or a visit I took to — or a time when I ate out at a restaurant and I'm to omit certain details. I omit the fact that I paid the bill in a restaurant, let's say or I finished the meal and then I went home. Still, you will tend to fill in the blanks. You'll tend to fill in the blanks with things you know. So, you might remember this later saying, "Okay. He told me he finished eating, paid the bill and left," because paying the bill is what you do in a restaurant.

This is benign enough. You fill in the blanks. You also can integrate suppositions made by others. And the clearest case of this is eyewitness testimony. And the best research on this has been done by Elizabeth Loftus who has done a series of studies, some discussed in the textbook, showing how people's memories can be swayed by leading questions. And it can be extremely subtle. In one experiment, the person was just asked in the course of a series of questions — shown a scene where there's a car accident and asked either, "Did you see a broken headlight?" or "Did you see the broken headlight?" The ‘the' presupposes that there was a broken headlight and in fact, the people told — asked, "Did you see the broken headlight?" later on are more likely to remember one. It creates an image and they fill it in.

In another study, she would show film segments and then ask, "Did you see the children getting on the school bus?" Now, there was no school bus but people who hear that question later on when asked, "Did you see a school bus in the film?" are more likely to say yes. In another study, she would show

people film segments and ask them either, "How fast were the cars going when they hit each other?" or "How fast were the cars going when they smashed into each other?" A week later she'd bring people back in to the laboratory and ask, "Did you see any broken glass?" Those who hear a smash tend to see the broken glass more than those who hear a hit because the question has changed their memory, making it more of a dramatic event.

Chapter 6. Hypnosis, Repressed Memory and Flashbulb Memories [00:44:56]

Hypnosis is the clearest case where there's a sort of reconstructive effort led by — led as a result of leading and probing questions. Some of you are readily hypnotizable and you can be hypnotized. And what we would learn about a past event from hypnotizing you will not necessarily then be inaccurate. What hypnotizing does is it makes people very willing to cooperate. Unfortunately, it isn't as if there is a memory storage there where you could just go through and look as in the movies where you just say, "What's the license plate?" The person's hypnotized and then the flashback comes in and then they zoom in on the license plate. Memory doesn't work that way. What happens is — What somebody will do in a hypnotizable state is they'll be very eager to please the hypnotist. And so they'll make stuff up.

And people under hypnosis just make stuff up. And they do very enthusiastically and very believably make stuff up. This is particularly the case with hypnotic regression when we ask you to go back to your sixth birthday party, for instance. And what's great as a developmental psychologist is if I ask you to go back to your fourth birthday party and you're hypnotizable you'll be oh, just like a four-year-old except you won't be like a real four-year-old. What you'll be like is an adult's notion of what a four-year-old is supposed to be. In fact, this has happened in the extreme case with hypnotic regression where people claim to speak languages like from ancient Egypt. And linguists love these studies because you don't — of course you don't really sound like you're speaking a language from ancient Egypt. What you sound is like a North American who believes he's speaking a language from ancient Egypt so they're, "nonsense sounds." [laughter] And so what it makes you is — Hypnotism brings out the actor in you. It makes you want to give a persuasive account of what happened. And so hypnotism is just an extreme form of what normally happens in eyewitness testimony.

Repressed memories. We could devote a class — We could devote a semester to the very heated debate in the United States mostly about repressed memories. There are many adults who have claimed to have experienced traumatic sexual abuse. In some cases, this is unexceptional from a memory point of view. People know this happened to them. They've always known it happens to them and then they tell people about it. But there's a subset of cases where people have had no memory up to a point of what happened to them. Then they go to a psychologist or a psychiatrist; they undergo questioning, often using hypnotic techniques; and then they recover a memory of past traumatic sexual abuse. And what this is — what makes this so debatable, and there is a debate about this. I don't want to try to preclude it one way or another. What makes this debatable is some psychologists believe that, in at least some cases, these memories are real and they have been repressed through a Freudian mechanism – that they're too terrible to bring to consciousness, and the therapy brings them out into real life. But most psychologists believe that these memories cannot be trusted, that these memories are created through the actions of the therapist.

And so, there's actually considerable psychological and legal battles over the veracity of the therapists where women who have claimed to have sexual — be sexually abused, for instance, have pressed criminal charges against their fathers on the basis of false memories. Similarly, people who have been accused of sexual abuse have pressed criminal charges against psychiatrists claiming that these psychiatrists have implanted the memories into their sons and daughters. It is controversial whether memories are ever repressed. What isn't controversial is that, for at least some cases, you can implant false memories in people, not because you're a sinister or evil person but because you really believed something happened. And you talked to them about it and then you caused these memories to come into being.

A final case is flashbulb memories. I asked this early in the semester. I'll ask it again. How many of you remember where you were on September 11, 2001? Is there anybody who doesn't remember where they were on September 11, 2001? It would be interesting. It was a socially relevant event, but here's the problem with these flashbulb memories. Flashbulb memories are the idea that these memories being so vivid, and they are vivid for many of us — exactly where we stood, what happened; well, they can't really be trusted. And here is why not. Because they are such important events, I bet many of you have actually heard the question before, "Where were you on September 11th?" and talked about it. What happens in these conversations is stories change. I have my — I knew where I was on September 11th. My wife knew where she was. But I spent as much time listening to her talk about it as I spent time me talking about it. And now maybe my memory is actually of her experience and not mine.

It's not — For all of these cases, the temptation you have to resist is saying, "Yes. I know memories can be swayed. I know they could be distorted and everything but, you see, I really am sure that happened." You have to resist that temptation because there are so many cases we know, including the tape of the girls that we just saw, where people are entirely sure things happened. And we know full well that they didn't exist. Being sure is no guarantee that a memory isn't false, reconstructed or even implanted.

So, this part of memory has three main morals. There are many types of memories. I talked about short-term memory, long-term memory. I talked about implicit memory and explicit memory. These are sort of separable sort of memories. You could break one while having the other one impaired. Arguably, there are brain systems dedicated to memory for faces, memory for everyday objects, memory for spatial locations. The key to remembering is organization and understanding. Introduction to "X" courses, including Introduction to Psychology courses, are among the hardest courses at Yale. And the reason why is there is just a lot of material that is diverse and you have to command each aspect separately. The easiest courses at Yale tend to be highfalutin seminars where you kind of have enough of a background that everything is — can be clear and understandable. The more you understand something, the easier you'll remember it.

And finally, you can't trust some of your memories. Your reading response for this week is you have to use your powers for good and not for evil, [laughter] though if you manage to succeed at this I will be very impressed. [laughter] But you have to describe, based on the lecture materials and the readings, how to implant a false memory.

Chapter 7. Question and Answer on Memory [00:52:47]

We have a few minutes. Any questions on memory. Yes.

Student: [inaudible]

Professor Paul Bloom: Uh huh. Hey. Please —

Student: [inaudible] Is that long-term sensory memory?

Professor Paul Bloom: The example is, "What sort of memory is it when you know how to play the piano?" And it's a very good question. It is long-term memory because you might know how to do a concerto or a song and then you have it stored in your head and you carry it around with you. You'll remember it a year from now, two years from now. It is long-term memory but it is also an excellent example of implicit memory because you know how to do it but you could do it unconsciously without attending to it. It's not sensory but it's as if, put it crudely, that your fingers know and not your mind. We have time for one more question. Yes.

Student: [inaudible]

Professor Paul Bloom: The question is about photographic memory. There are a lot of claims about photographic memory. My understanding is they do not tend to be substantiated. Sometimes photographic memory, and this came up when we talked about autism a few classes ago, is linked with savant-like skills. People who have severe impairments in some ways may have photographic memories in others. I am not convinced that photographic memory in the sense that you see something, you take a picture of it, you hold it in memory really exists. I think there may be one or two case studies that suggest it might be real but I think it's controversial. Okay. We have a guest lecturer on Wednesday. Dean Peter Salovey will talk to us about love.